

which is synchronised three-phase, offset by 120-degree, AC power of the order of thousand volts at the line rate of 50 cycles per second. Three phases are chosen to get nearly peak value at any given instant, which results in a good compromise between cost and performance. More phases could be used but this implies more cost with only a slight improvement in performance.

The AC power generated is now ready for its journey to the customers. As mentioned earlier, a three-tiered hierarchy is used to transmit this AC power to distant end-users. Power transferred over lines is given by the product of voltage and carrying current.

For a given line resistance, which depends on the line material and line length, the power loss is given by the product of line resistance and square of the carrying current. So in order to reduce power loss in the lines and transfer maximum generated power to long-distant end-users, current must be made as small as possible and voltage as large as possible. That's why transmission substations located next to power plants use large transformers to step up generator output voltage, thus allowing megawatts of power transmission over distances of 400 km or more.

At power substations, voltages are stepped down and lines branched out to cover larger areas. This is performed successively, transforming and branching out from extremely high voltage (typically 345 to 785 kV) to HV (typically 115 to 230 kV), and then from HV to MV (typically 2.4 to 69 kV), and finally from MV to LV (typically 120V to 600V) for delivery to homes or offices.

HV lines transmit power over distances that are tens of kilometres, but are very noisy and not used to carry broadband signals. Such a high voltage contains infinite number of harmonic components, and if it spikes at the frequency of the RF used to transmit data, it will cancel out that signal and the data transmission will be dropped or damaged en-route.

MV lines are capable of transmitting power over a few kilometres, while LV lines cover only a few hun-

dred metres. MV and LV lines are preferred for use in BPL data transmission because of their low noise level.

In order to use MV/LV lines for broadband services, BPL systems are designed to operate in the frequency spectrum range of 2 to 80 MHz. BPL signal may be injected onto power lines between two phase conductors, between a phase conductor and the neutral conductor, or on a single phase or neutral conductor. The BPL signals are injected into and extracted from MV/LV lines through inductive or capacitive couplers.

Injector is a device that aggregates user data onto the power lines and provides an interface between power line and BPL operation support system that is connected to the Internet backbone. Concentrator provides an interface between power lines carrying the BPL signal and the households.

In inductive coupling, BPL signal couples onto the line by wrapping inductors around the line. On the other hand, capacitive coupling uses a capacitor for coupling and the signal is modulated onto the line voltage. Inductive couplers are known to be rather lossy, but since they require no physical connection to the network, they are safer to install on energised lines than capacitive couplers.

Since HF signals have the rare ability to support long-distance point-to-point communications due to line noise, signal attenuation and limitations on the amount of signal power that can be injected onto power lines without causing unreasonable interference for other spectrum users, repeaters are required in between the transmission and reception ends. This can be done by using MV couplers to couple the broadband signal off the MV line so that it can be regenerated, if necessary, and amplified before being fed back onto the MV line through another coupler.

The distribution transformers that change voltage levels between MV and LV lines are particularly harsh on the weak broadband signal. These transformers, due to their high inductance, act as low-pass filters and appear as open circuits for the passage of high-frequency BPL signals. This implies

BPL can be deployed either as end-to-end BPL or as hybrid BPL. The end-to-end BPL system uses both the access BPL and the in-house BPL while in hybrid BPL, the bypass box does not couple the broadband signal to/from the LV line but converts it to/from a wireless format and delivers it to the wireless access point

that BPL signals going between MV and LV lines need to bypass the transformers. Typically, a bypass box can be used, which may consist of coupler and built-in repeating functionality.

Connectivity to the backhaul network (Internet, PSTN, UMTS, 2G, 3G, 4G, CDMA, WiMAX, etc) is provided through the operation support system (OSS) coupled to an MV distribution line. The OSS converts data formats, aggregates and concentrates uplink data streams, provides routing functionality, helps allocate bandwidth and resources, generates billing and charging data, and provides various backhaul Ethernet interfaces to fibre-optic or wireless connections. It consists of various servers like authentication and authorisation server, dynamic host control protocol server, domain name system server and billing server to perform the required back-end tasks.

BPL can be deployed either as end-to-end BPL or as hybrid BPL. The end-to-end BPL system uses both the access BPL and the in-house BPL, i.e., power lines are used all the way from the power substation to the end user. In this case, BPL signal can either bypass the MV/LV transformer or go through the transformer (Fig. 1).

In hybrid BPL, the bypass box does not couple the broadband signal to/from the LV line but converts it to/from a wireless format and delivers